Y-axis and similarly the term $\sum my^2$ is the moment of inertia about X-axis. Thus,

$$I_X = \sum my^2 \quad \text{and} \quad I_Y = \sum mx^2$$

Substituting in the equation for I_z gives,

$$I_Z = I_X + I_Y$$

Thus, the perpendicular axis theorem is proved.

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EXAMPLE 5.16

Find the moment of inertia of a disc of mass 3 kg and radius 50 cm about the following axes.

(i) axis passing through the centre and perpendicular to the plane of the disc,

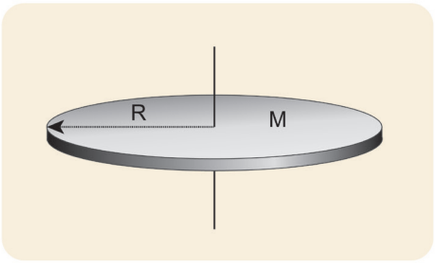
(ii) axis touching the edge and perpendicular to the plane of the disc and

(iii) axis passing through the centre and lying on the plane of the disc.

Solution

The mass, M = 3 kg, radius R = 50 cm = 50×10^{-2} m = 0.5 m

(i) The moment of inertia (I) about an axis passing through the centre and perpendicular to the plane of the disc is,

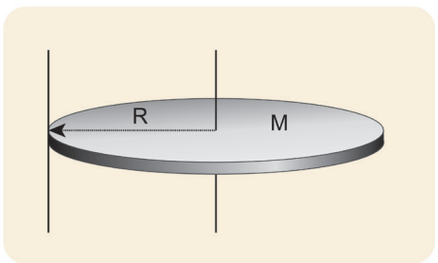


$$I = \frac{1}{2}MR^2$$

$$I = \frac{1}{2} \times 3 \times (0.5)^2 = 0.5 \times 3 \times 0.5 \times 0.5$$

$$I = 0.375\,\text{kg m}^2$$

(ii) The moment of inertia (I) about an axis touching the edge and perpendicular to the plane of the disc by parallel axis theorem is,



$$I = I_C + Md^2$$

where, $I_C = \frac{1}{2}MR^2$ and $d = R$

$$I = \frac{1}{2}MR^2 + MR^2 = \frac{3}{2}MR^2$$

$$I = \frac{3}{2} \times 3 \times (0.5)^2 = 1.5 \times 3 \times 0.5 \times 0.5$$

$$I = 1.125\,\text{kg m}^2$$

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